INCH-POUND

MIL-PRF-83363F
20 November 2024

SUPERSEDING
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PERFORMANCE SPECIFICATION

GREASE, TRANSMISSION, HELICOPTER (NATO G-396)

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of extreme pressure, anti-wear helicopter transmission grease. This grease is identified by NATO Code Number G-396 (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791	Testing Method of Lubricants, Liquid Fuels, and Related Products

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D217	Standard Test Methods for Cone Penetration of Lubricating Grease
ASTM D972	Standard Test Method for Evaporation Loss of Lubricating Greases and Oils
ASTM D1478	Standard Test Method for Low-Temperature Torque of Ball Bearing Grease
ASTM D2266	Standard Test Method for Wear Preventative Characteristics of Lubricating Grease (Four-Ball Method)
ASTM D2596	Standard Test Method for Measurement of Extreme-Pressure Properties of Lubricating Grease (Four-Ball Method)
ASTM D4057	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D6184	Standard Test Method for Oil Separation from Lubricating Grease (Conical Sieve Method)

(Copies of these documents are available from https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Toxicity. The materials shall have no adverse effect on the health of personnel when used for their intended purpose.

3.2 Materials. The physical composition of the grease shall be such that the material passes the requirements in section 3. The grease shall be a homogenous mixture of base fluids, thickener(s), and additives as necessary. A formulation of polyol aliphatic ester, a fluorinated polysiloxane, a sodium montmorillonite clay, fluorinated ethylene propylene copolymer, antimony dialkyldithiocarbamate, and other additives as necessary is recommended but not mandatory.

3.3 Finished grease. The properties of the finished grease shall be as specified in 3.3 and Table I when tested as specified in section 4.

3.3.1 Unworked penetration. The unworked penetration (as measured in units of 0.1 mm) of the grease shall be a maximum of 340 when tested as specified in Table I.

3.3.2 Worked penetration. The worked penetration (as measured in units of 0.1 mm) of the grease shall be a maximum of 340 when tested as specified in Table I.

3.3.3 Prolonged worked penetration. The penetration (as measured in units of 0.1 mm) of the grease after working 100,000 double strokes shall be a maximum of 375 when tested as specified in Table I.

3.3.4 Evaporation. The grease shall not lose more than 5 percent of its weight after 22 hours at 300 °F (148.9 °C) when tested as specified in Table I.

3.3.5 Oil separation. The grease shall not lose more than 10 percent of its weight after 30 hours at 300 °F (148.9 °C) when tested as specified in Table I.

3.3.6 Load wear index. The load wear index value of the grease shall be not less than 90 when tested for load carrying capacity as specified in Table I.

3.3.7 Wear preventative characteristics. When tested as specified in Table I the wear scar diameters shall not be greater than those listed in Table I. The temperature shall be monitored and recorded for the ambient temperature, four-hour test and the temperature shall be regulated at 167 °F (75 °C) for the two-hour test.

3.3.8 Low temperature torque. A number 6204 open ball bearing lubricated with grease shall give a starting torque of not more than 3,000 g-cm and a running torque of not more than 1,000 g-cm when tested as specified in Table I.

3.3.9 Storage stability. After storing for six months as specified in 4.4.2, the worked penetration (as measured in units of 0.1 mm) of the grease shall not have changed from the original worked penetration by more than 30 units. The unworked penetration shall be a minimum of 250 (as measured in units of 0.1 mm).

3.3.10 Workmanship. The grease shall be a high quality, smooth, homogenous product, free from abrasives and impurities when examined as specified in 4.4.3.

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3.4 Recycled, recovered, environmentally preferable or biobased materials. Recycled, recovered, environmentally preferable or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

TABLE I. Properties of the finished grease.

Property	Test Limits	Test Methods		
		Section	FED-STD 791	ASTM
Workmanship	Pass	4.4.3		
Cone penetration				D217
Unworked penetration, 0.1 mm	max 340			
Worked penetration (60 double strokes), 0.1 mm	max 340			
Prolonged worked penetration (100,000 double strokes), 0.1 mm	max 375		313	D217[1]
Evaporation loss, 300 °F (148.9 °C) for 22 hours, weight percent	max 5		350	D972[1]
Oil separation, 300 °F (148.9 °C) for 30 hours, weight percent	max 10			D6184
Load-wear index, kgf	min 90			D2596
Wear preventative characteristics, wear scar average, mm				D2266
AISI-E-52100 steel, 167 °F (75 °C) for 2 hours, mm	max 1.30			
AISI-E-52100 steel[2], ambient temperature for 4 hours, mm	max 1.30			
Low temperature torque of ball bearing 54 °C ± 1 °C start/run @ 1 hour				D1478
Starting torque, g-cm	max 3,000			
Running torque, g-cm	max 1,000			
Storage stability, 100 °F ± 5 °F (38 °C ± 3 °C) for 6 months[3]		4.4.2		
Unworked penetration, 0.1 mm	min 250			D217
Worked penetration (60 double strokes), change from original, 0.1 mm	max 30			D217

NOTES

1.	Referee Method.
2.	The temperature shall be monitored and recorded for the ambient temperature, four-hour test.
3.	The sample of grease will be stored in a 1-pound (0.454 kg) oil tight container.

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4. VERIFICATION

4.1 Classification of inspection. The examination and testing of the grease shall be classified as conformance inspection. (see 4.1.1).

4.1.1 Conformance inspection. The conformance inspection shall consist of tests from Table I for workmanship, worked penetration, evaporation loss, and wear preventative characteristics (wear scar average) at 167 °F (75 °C) for 2 hours. Samples shall be labeled completely with information that identifies the purposes of the sample, name of product, specification number, lot/batch number (see 6.4), date of manufacture, and date of sampling.

4.2 Sampling. Sampling shall be in accordance with ASTM D4057.

4.3 Method of inspection. Tests shall be performed in accordance with the methods specified in Table I, 4.4.2 through 4.4.3, and the contractual requirements (see 6.2). Physical and chemical values apply to the average of the determinations made on the samples.

4.3.1 Inspection. Unless otherwise specified, inspection testing shall be performed in accordance with 4.3.2 and the contractual requirements (see 6.2).

4.3.2 Grease. The grease shall conform to the requirements for finished grease that shall be determined by appropriate examination and testing in accordance with 4.3.3.

4.3.3 Physical and chemical values. Tests shall be performed in accordance with the applicable methods specified in Table I and 4.4.2 through 4.4.3. Physical and chemical values specified in section 3 apply to the arithmetic average of the determinations made on the samples for those values that fall within any stated repeatability or reproducibility limits of the applicable test method.

4.4 Test procedures.

4.4.1 List of test methods. The tests of this specification shall be conducted in accordance with the applicable test method as listed in Table I and other methods as described in 4.4.2 and 4.4.3.

4.4.2 Storage stability. The penetration after storage shall be determined on a sample of grease which has been stored at a temperature of 100 °F ± 5 °F (38 °C ± 3 °C) in a 1-pound (0.454 kg) oil tight container for 6 months subsequent to the original determination of the penetration. The worked and unworked penetrations after storage shall be determined in accordance with ASTM D217.

4.4.3 Examination of the product. The sample grease shall be inspected visually and a suitable portion shall be worked with a spatula on a glass surface. After working, the grease shall be spread with a straight edge and observed for uniformity as indicated by a smooth surface.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's System Commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The grease covered by the specification is intended for use in military helicopter tail rotors, intermediate transmissions, and gear boxes. Other applications include screw jack actuators, fine pitch gear trains, servomechanisms, journal bearings, and helicopter rotor hubs.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required.

c. Packaging requirements (see section 5.1).

6.3 International standardization agreement implementation. This specification implements:

a. NATO STANAG/AFLP-1135 *Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations*.

When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the International Standardization Agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.4 Safety Data Sheets. Contracting officers will identify those activities that require copies of completed Safety Data Sheets (SDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

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6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume 1, *Shelf-Life Management Program: Program Administration*; and Volume 2, *Shelf-Life Management Program: Materiel Quality Control Storage Standards*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.6 Subject term (key word) listing.

Extreme pressure grease
Anti-wear grease
Tail rotor
Gear box
Intermediate transmissions
Screw jack actuators
Fine pitch gear trains
Servomechanisms
Journal bearings

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
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Navy – AS
Air Force – 68
DLA – GS

Review activities:
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Air Force – 11

Preparing activity:
Air Force – 68
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